Padraig Lavery

Software Engineer – LLM Applications, RAG & Agents | Python, Node.js, AWS

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Software engineer with 4+ years building and running production systems for UK and Middle East banking clients on AWS and Kubernetes, now building LLM applications. Hands-on with retrieval-augmented generation, agentic tool-calling, pgvector and local models through Incident Copilot, plus earlier ML work on fraud detection and trading. Brings production discipline to AI work: safe failure, testing, security and runbooks. Preparing for the AWS Certified AI Practitioner exam (November 2026).

AI AND ML PROJECTS

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama, Docker, GitHub Actions

- Incident-triage service: takes a production incident description, retrieves relevant runbook content and suggests likely root causes and next steps, using agentic tool-calling for live service status.
- Moved inference from AWS Bedrock to locally hosted models via Ollama to remove inference costs. Runbook knowledge base and FastAPI service in place; retrieval, model integration and testing in development.

ML Trading System (in progress) | Python, Alpaca API

- Built a backtester for SEC Form 4 insider purchases by company CEOs, then extended it into an ML-driven strategy running on a paper-trading account, focused on low-frequency, high-conviction trades.

AI-Based Credit Card Fraud Detection (final year project) | Angular, Flask, GraphQL, Docker, Kubernetes, AWS SageMaker

- End-to-end web app flagging irregular credit card transactions as Docker/Kubernetes microservices on AWS, with a model trained in R and Python and served via SageMaker.

Football Player Performance Model | Python, soccerdata

- Notebook model predicting how player performance carries over across leagues and teams, using fbref and Transfermarkt data.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

- Built the CI/CD pipeline for a UI Builder service as part of an internal AI investigation project.
- Eliminated recurring daily out-of-memory crashes in a high-volume transaction ingestion service by loading caches once per batch instead of per file; mitigated memory-exhaustion failures in ETL jobs.
- Redesigned a month-end invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation.
- Designed and implemented audit logging of staff access to customer accounts across UK and Middle East bank portals.
- Hardened authentication and account services and remediated critical Snyk-flagged vulnerabilities for PCI DSS compliance.
- Led a cross-team initiative for automated API testing, including OpenAPI documentation, unit tests and stabilised GitHub Actions CI.
- Authored runbooks, SOPs and technical guides; mentored four new starters and run the team's retrospectives.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a cloud-cost web API using Go, Docker, Kubernetes and Git.

SKILLS

- **AI and ML:** LLM integration, retrieval-augmented generation, agentic tool-calling, embeddings with pgvector, Ollama, AWS Bedrock, SageMaker
- **Languages:** Python, JavaScript (Node.js), C#, Go, SQL, R
- **Back end:** FastAPI, Flask, Node.js, .NET Core, GraphQL, OpenAPI
- **Data:** PostgreSQL (incl. pgvector), MongoDB
- **Cloud and delivery:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Docker, Terraform, GitHub Actions

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026